



GAME-SPECIFIC FACULTY GUIDE

Labor Force Files

A labor-market measurement game built around classification, unemployment, participation, and worker flows.

Use Labor Force Files to connect labor-market rates to the people counted in them and explore what a change in unemployment tells us.

Why This Game Exists

I wanted students to decide where people belong before treating labor-market statistics as formulas. They can calculate an unemployment rate correctly and still be unsure who counts as unemployed or why the rate changed. Classifying people first gives the arithmetic something to describe.

I also thought about where students encounter these numbers: reports and newspaper headlines. That led to the newspaper-report feel and film-noir look. Students build the labor force, calculate the rates, and follow people moving between groups. I wanted them to look past the headline and explain what happened to the people behind it.

What Students Do

The game has nine stages and 17 checks: calculations, four person classifications, predictions, and interpretations. Each untimed run selects one of five baselines and a set of change cases, keeping the adult population fixed.

Students begin by building the labor force from exact employment and unemployment counts. They then classify four people whose situations involve part-time work, active job search, retirement without search, and discouragement. The next two stages ask them to calculate the baseline unemployment rate (UR) and labor-force participation rate (LFPR). The six crowd illustrations support the groupings; students use the report's exact counts and never need to count silhouettes.

File 5 presents a direct change: either unemployed people find jobs or employed people lose jobs and begin searching. Students predict UR, calculate it, and identify what happens to participation. File 6 introduces new participants, asking for a prediction, both rates, and an explanation of how employment and UR can rise together. File 7 follows discouraged workers out of the labor force. File 8 combines job finding with entry, requiring a new labor-force count, both rates, and an interpretation.

Every later case independently compares with the same baseline; one file's results do not carry into the next. File 9's headline audit asks students to select the population-flow explanation supported by the before-and-after counts and both rates.

Feedback allows retries and identifies fields needing correction while retaining correct entries. On rate calculations, **NEED A HINT?** gives denominator guidance; **SHOW FORMULA** separately reveals the formula and its spoken equivalent. Students can attempt retrieval first. Hints carry no score or mastery penalty.

Results show overall first-attempt accuracy, UR and LFPR checks, classification performance, headline completion and attempts, and a takeaway. Rate summaries count complete checks, including those with several fields. Its Labor market through the files graph and table use the student's actual cases: baseline (Files 1–4), direct change, new participants, discouraged workers, mixed flows, and headline audit. The graph connects these six points in file order for comparison, not as a time series. Its two panels have different labeled percentage scales; the table supports exact comparisons.

What the Game Is Really Teaching

The four people give students a reason to sort out employment status. In this simplified model, having a job counts as employment, including part-time work. An unemployed person has no job, is available for work, and actively searches. Someone without employment or active search is outside the labor force. Wanting a job alone does not establish unemployment status.

Those classifications let students build the accounts: Labor Force = Employed + Unemployed. The remaining adults are outside it: Not in Labor Force = Adult Population – Labor Force. The adult population includes both groups and is larger than the labor force in the game's cases. Ask students to locate each person within that structure before interpreting a rate.

They calculate $UR = (Unemployed \div Labor\ Force) \times 100$ to measure the unemployed share of the labor force. People outside the labor force are excluded from its denominator. A change in UR can therefore reflect a change in unemployment, a change in the labor force, or both.

For participation, $LFPR = (Labor\ Force \div Adult\ Population) \times 100$ measures how much of the adult population is working or actively searching while available for work. Its denominator includes adults outside the labor force. The two rates answer different questions, which helps explain why they can move differently.

When an unemployed person finds work, employment rises and unemployment falls within the same labor force. Job loss followed by active search reverses that movement. Entry from outside the labor force adds to employment or unemployment and raises participation. UR then depends on the mix of working and searching entrants: it can rise, fall, or stay unchanged. The mixed cases require students to account for job finding and entry together.

Later, students meet discouraged workers: people who want and are available for work but have stopped actively searching because they believe suitable work is unavailable. In this model, they are outside the labor force. When unemployed workers become discouraged, unemployment and the labor force fall by the same number while employment stays unchanged. Both UR and LFPR fall in the game's cases. The lower UR accompanies withdrawal from search, with no increase in jobs; the definition depends on search status and its reason, not the length of unemployment.

By the headline audit, students have seen how employment can grow while UR rises if enough new participants are still searching. Conversely, a lower UR can accompany discouragement. Ask students to connect both rates to the counts and the underlying flows before deciding what a headline supports. The game uses controlled comparisons in a simplified model; official labor-market measurement involves more detail.

The Educational Loop

Students work with population groups and observe changes in counts and rates. Feedback explains classifications, denominators, and flows, while calculations formalize these relationships. Mixed flows and the headline audit supply transfer within the activity by combining several measures in a new case. The final report supports discussion of their reasoning.

Experience → Consequence → Economic Explanation → Formalization → Transfer

Using It in Class

These are possible starting points; choose what fits your course.

Before formal labor-market measurement instruction

Use a brief classification discussion before opening the game. Then let students build the labor force from the report, examine the four people, and develop the rate formulas from those categories.

During instruction

Use a run after introducing employed, unemployed, labor force, and outside the labor force. Pause before calculations and ask which denominator belongs in each rate and what population it represents.

After instruction

Use the later files for application. Ask students to explain why a rate changed by tracing the people who moved between categories. Return to the baseline for each comparison.

Independent/student review

Students can use feedback, optional hints, corrections, and the final report independently. You can request a short reflection or results screenshot through your usual course workflow; the game sends no automatic faculty report.

Questions to Consider

Select any of these prompts for discussion or a short reflection.

1. Why does wanting more hours leave a part-time worker in the employed group?
2. What information separates unemployment from being outside the labor force in this model?
3. Why do UR and LFPR use different denominators? What does each denominator let us measure?
4. When unemployed people find jobs, why can UR fall while LFPR stays unchanged?
5. How can employment rise at the same time as UR? Trace what new participants are doing.
6. Why can discouragement lower UR without creating jobs? What does LFPR add to that interpretation?
7. What would you check before accepting a headline that treats a lower UR as evidence of more jobs?
8. How should you read the final graph differently from a graph of consecutive months?

Common Misconceptions

- “Anyone without a job is unemployed.” Availability and active search matter. Discouraged people who have stopped searching are outside the labor force in this model.
- “Part-time workers are unemployed.” Having a job counts as employment even when the person wants more hours.
- “UR divides unemployment by the adult population.” Its denominator is the labor force; LFPR uses the adult population.
- “LFPR is the share of the labor force that is working.” It measures the share of adults who are employed or unemployed.
- “A lower UR proves that people found jobs.” Search withdrawal can lower UR too. Check employment, participation, and the flows.
- “The files form a time series.” Every later case compares independently with the same baseline. The connecting lines show file order.